



UGANDA CHRISTIAN
UNIVERSITY

A Centre of Excellence in the Heart of Africa

Department of Computing & Technology

*INCORPORATION OF REINFORCEMENT LEARNING AND
EXPLAINABLE(XAI) IN HUMAN-IN-THE-LOOP
SOFTWARE TO ASSESS PRIORITIZATION, DIAGNOSIS,
AND TREATMENT IN DERMATOLOGY.*

By

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A proposal submitted to the Faculty of Engineering Design &
Technology for the study leading to the Degree in partial fulfilment of
the requirements for the award of **Bachelor of Science in
Computer Science** of Uganda Christian University

SPECIALIZATION TRACK: *Artificial Intelligence and Machine
Learning*

Monday, 16th March, 2026.

Declaration

I, *Mutegeki James*, declare to the best of my knowledge that the work in this proposal is original and has not been submitted before to any university/institution. It is the result of my effort.

A handwritten signature in black ink, appearing to be 'Mutegeki James', written in a cursive style with a horizontal line through the middle.

Signature: _____

Date: 25/03/2026

Ethical Generative AI Usage Declaration

Students may use generative artificial intelligence (AI) ethically as a “study buddy” or assistant rather than a replacement for critical thinking. This includes citing AI use and fact-checking all outputs to maintain academic integrity.

For purposes of this assignment, the following shall guidelines shall apply.

- i. Generative AI may be used for brainstorming, and generating ideas to improve your work. You may also use AI for literature search, and summarizing journal papers for quick analysis and assessing relevance of publications. However, once the relevant publications have been identified, a student is expected to fully review the selected publications, identifying problems addressed, methods used, gaps identified and recommendations made.
- ii. Where AI has been used as guided in (i) above, full disclosure of how it has been used should be made in the appendix C, including how AI was used, the AI tools (including mode versions), and a compilation on major AI prompts used.
- iii. NO GENERATIVE AI CONTENT IS ALLOWED IN THE FINAL SUBMISSION. The final submission SHOULD be your original work.

A. Declaring the non-use of AI for content generation

I, *Mutegeki James*, declare that no content generated by AI technologies has been used in this proposal.



Signature: _____

Date: 25/03/2026

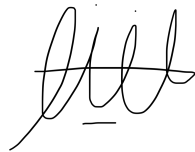
B. Acknowledging the use of AI as a study buddy

Please refer to Appendix C for the complete declaration of AI usage for generating ideas, brainstorming, literature search, and summarizing only to improve your work.

Approval

This proposal has been submitted for examination with approval of the following advisors/supervisor.

Department of Computing & Technology
Faculty of Engineering Design & Technology
Uganda Christian University

A handwritten signature in black ink, consisting of stylized, overlapping loops and a horizontal line across the middle.

Signature: _____

Date: 25/03/2026

Abstract

This research study aims to develop a Human-in-the-Loop system for patient diagnosis and triage. The research study aims to solve two problems. One such problem is the lack of dermatologists in Uganda. Another problem is the phenotypic bias shown by the current AI system. AI has shown potential in the diagnosis of skin diseases. However, the current AI system has phenotypic bias. This means that the current system performs well for people with lighter skin tones but less well for people with darker skin tones. The current system does not provide transparent information. This is a problem for the local dermatologists who want to use this system and know how it comes to these conclusions. The proposed solution combines Reinforcement Learning (RL) and Explainable AI (XAI) in the form of a collaborative interface. Contrary to other static approaches, the RL-based agent uses a reward function that prioritizes fairness by heavily penalizing the misdiagnosis of pigmented skin and strives for accurate and equitable diagnosis for all Fitzpatrick skin types. In an effort to address the transparency issue, the proposed solution uses Concept Bottleneck Models and salience maps to interpret the complex decisions made by AI. The local medical officers use this interface to conduct audits of AI system logic while they deliver immediate feedback. The Human-In-The-Loop system uses active learning to develop the AI system, which transforms the software from a basic conclusion marker into a diagnostic partner who works with users. The research seeks to develop a system that meets cultural needs and physiological requirements, which will enable medical professionals to assess life-threatening conditions like melanoma and Kaposi Sarcoma in areas with limited resources.

I dedicate this research to my beloved parents, whose unwavering support and belief in the power of education have been the foundation of my academic journey at Uganda Christian University. Your sacrifices and encouragement provided me with the necessary strength to complete my Bachelor of Science degree in Computer Science and to create healthcare solutions that would benefit our community. Your constant support brings me both motivation and happiness. I hope this work serves as an inspiration for you to pursue your own dreams with the same passion and dedication, knowing that through innovation and technology, we can create a more equitable future.

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I want to thank all the people who helped me create this research proposal (Incorporation of Reinforcement Learning and Explainable (XAI) in Human-in-the-Loop Software to Assess Prioritization, Diagnosis, and Treatment in Dermatology). My first priority goes to my lecturers at Uganda Christian University because of the hard work and effort input into the Department of Computing to educate us on different matters and sectors of technology. Through their academic guidance and technical knowledge of Artificial Intelligence they helped me evolve into a researcher who uses LaTeX and advanced research methods to achieve professional excellence. Through their constant support and belief in my abilities my parents built the essential support system which drives me forward in my academic journey. Your dedication and support have enabled me to study for a Bachelor of Science in Computer Science and work on projects which seek to enhance healthcare services in our nation. I want to show my appreciation to my friends and colleagues who have supported me during my academic journey. Through our discussions and peer reviews and our late-night study sessions we created an environment which helped me understand complicated aspects of Reinforcement Learning and Human-in-the-Loop systems.

Chapter 1

Introduction

1.1 Background

1.2 Problem Statement

The dermatological AI system is faced with three major challenges, all of which stem from the following three challenges. The first problem is the need for algorithms that identify the condition of the skin, and currently, the algorithms only work with Caucasian tones. The second problem is the fact that most of the data is based on the International Skin Imaging Collaboration (ISIC) archive, and the data on the tone of the skin is based on data that reflects lighter tones of the skin, which fall in the Fitts Patricks (I, II, III) category. The second problem is the fact that the medical officers need an explainability component in order to understand whether the flag in the AI system is a result of pathology or background noise. The third problem with the traditional models is the fact that they cannot adapt to the local patterns of disease spread, including HIV-related Kaposi Sarcoma.

1.3 Research Objectives

The research aims to create and assess a Human-In-The-Loop software system which combines Equity-Driven Reinforcement Learning with Explainable AI to enhance the fairness, transparency, and accuracy of dermatological triage and diagnosis and treatment processes for Ugandan patients with various skin tones.

1.3.1 Specific Objective 1(Explainability Integration)

To integrate clinical explainability into the system using Concept Bottleneck Models (CBMs) that translate AI deductions into standardized medical terms.

1.3.2 Specific Objective 2(Clinical Validation)

To deploy a Human-In-The-Loop system interface tailored for Ugandan health centers, allowing local dermatologists to retrain the model through daily diagnostic overrides or feedback.

1.3.3 Specific Objective 3(Technical Development)

To engineer a Reinforcement Learning-based triage agent using Proximal Policy Optimisation that utilizes a weighted penalty function to eliminate the accuracy gap between light and dark skin tones.

1.4 Justification

The research needs to proceed because Uganda's healthcare system faces specific inequalities and the medical system lacks technological resources to treat dermatological conditions. Skin diseases cause more than half of all outpatient visits which do occur in Uganda so the country requires more dermatology specialists to treat this patient needs. The research creates a clinical decision support tool which enables non-specialist health workers to perform specialized diagnostic work.

The existing dermatological AI systems show major shortcomings because they fail to recognize different skin tones which hinders their ability to diagnose skin conditions that affect people with darker skin types from Fitzpatrick IV to VI range. The researchers use a fairness-weighted reward function to prove their chosen method as their research approach which supports the just treatment of local people.

Deep learning models maintain their opaque nature because they generate probability scores which do not offer clinical explanations to medical personnel who need to trust these systems. Dermatologists can use Explainable AI and Concept Bottleneck Models to verify AI decisions through biological features without needing to use image analysis.

Chapter 2

Literature Review

Artificial Intelligence (AI) development progressed from theoretical research to practical application in clinical dermatology which revealed multiple systemic challenges that need resolution for complete operational success. Clinical outcomes display a primary concern because different skin types produce different treatment results according to medical research which shows that skin cancer occurs less often in darker skin populations than in Caucasian groups yet brings greater health risks [1]. Ugandan medical practice requires both doctors and artificial intelligence systems to obtain specialized training which enables them to identify skin diseases in people with "skin of color" for better early disease detection and survival outcomes [1]. Mulago National Referral Hospital research demonstrates how complicated the local dermatological situation has become because of the combined effects of skin disorders and systemic illnesses. The hospital STD clinics recorded historical data which showed that 43 percent of patients with genital ulcers had HIV according to their recent sexually transmitted disease history [2]. The research demonstrates the need for diagnostic solutions which must consider the Ugandan disease profile because skin disorders now represent the fourth most common cause of non-fatal disease worldwide according to current data [3]. Researchers need to validate their findings through real-world testing which requires performance assessment methods that maintain their effectiveness when applied to various demographic groups including those who are typically underrepresented in research studies [4]. From a technical perspective, the implementation of High-Resolution (HRNet) architectures is vital for achieving the pixel-level precision required for advanced visualization techniques [5]. These architectures facilitate the generation of Saliency Maps and Integrated Gradients, which serve as an "explanation layer." This layer is critical for clinical auditability, allowing medical officers to verify that the AI is focusing on actual pathological features rather than confounding artifacts like hair or varied lighting [6]. Furthermore, the evolution of AI into a "diagnostic partner" rather than a standalone tool has been shown to drastically reduce the mismanagement of malignant lesions—dropping error rates from 58.8 percent to a mere 4.1 percent [7]. Finally, achieving ethical and safe clinical implementation requires a move toward customized medical systems that prioritize data representativeness [8]. By optimizing Convolutional Neural Networks (CNNs) for varied skin textures, it becomes technically feasible to identify complex lesions such as Kaposi Sarcoma, which is highly prevalent in the region [9]. This focus on Fitzpatrick IV-VI skin tones ensures that the software moves beyond simple diagnosis into a collaborative intelligence framework that effectively manages treatment prioritization and triage within the broader Ugandan healthcare system [10].

The research needs High-Resolution (HRNet) architectures because they provide the essential pixel accuracy needed for their advanced visualization studies [5]. The systems produce

Saliency Maps together with Integrated Gradients which serve as an "explanation layer." This layer enables clinical auditing because medical officers can check whether AI systems detect real pathological features or exclude non-existent conditions that include hair and lighting alterations [6]. The development of AI into a "diagnostic partner" has demonstrated a major decrease in malignant lesion mismanagement because it reduced error rates from 58.8 percent to 4.1 percent [7]. The development of ethical and safe medical practices requires a shift towards personalized healthcare systems which maintain accurate data representation [8]. Technical systems for skin texture classification enable the detection of complex dermatological conditions which include Kaposi Sarcoma, because it occurs frequently in that geographic area [9]. The software development team used Fitzpatrick IV-VI skin tones as their primary research focus to build a software system that enables users to work together on treatment selection and patient assessment throughout the entire Ugandan healthcare network [10].

Self-Supervised Learning (SSL) together with dark skin tones from Fitzpatrick IV-VI skin types represents the solution for this existing data scarcity problem. SSL enables models to extract skin structure elements from extensive unlabeled datasets yet requires additional training on specific Ugandan clinical datasets that include high-quality data [11]. These systems function as collaborative intelligence systems which enhance clinician work through their ability to identify high-risk patients who require urgent treatment in settings with limited medical resources [12]. The path to automated dermatological diagnosis has transformed its focus from basic classification methods toward architectural analysis. The development of deep learning research in dermatology used Residual Networks (ResNet) because these models achieved exceptional performance yet did not meet the transparency requirements needed in clinical settings at Mulago National Referral Hospital [13]. High-Resolution (HRNet) frameworks currently represent the most advanced technology because these systems preserve high-resolution data throughout their execution without needing to retrieve it from low-resolution components. The system detects subtle pigment changes by preserving pixel-level details, which permit the detection of hypo-pigmentation and hyper-pigmentation across various Kaposi Sarcoma stages and different HIV-related skin conditions that affect patients with darker skin types [14]. Attention mechanisms have brought about a complete transformation in model performance because these systems now excel at handling glitches, which include dense terminal hair and surgical markers that commonly cause incorrect results in uncontrolled clinical environments. The model uses Self-Attention layers to determine which spatial regions matter most, while it automatically ignores all non-pathological visual elements [15]. The development of this technology enables the creation of Grad-CAM (Gradient-weighted Class Activation Mapping), which produces a visual heatmap that shows how the AI makes its decisions to doctors. The technical system provides essential transparency, which enables doctors to use the model because it shows the AI's focus through saliency maps, which medical officers can use to decide whether to check hair areas or actual skin lesions[16].

Chapter 3

Methodology

This research study reveals a mixed method technique to solve clinical dermatology challenges in the clinical setting of Uganda. The research develops a Human-in-the-Loop (HITL) software interface that enables automated AI systems to function under expert clinical supervision. The triage system uses Proximal Policy Optimization (PPO) as its main algorithm. The RL agent uses the diagnostic process as a dynamic system that makes multiple decisions which produce results that can either be negative or positive based on actual clinical results. The model implements a weighted penalty matrix to handle phenotypic bias through its Equity-Driven Reward Function. The reward function presents a design which increases penalties for false negative results that affect Fitzpatrick Type VI skin patients who have darker skin tones. The primary security requirement for the model demands that it must achieve complete skin-tone invariance. The system uses two XAI main components to establish transparency while minimizing "automation bias" through its design. The Concept Bottleneck Models (CBM) force AI systems to recognize actual dermoscopic features which they must use to express their findings in medical standards rather than providing simple probability assessments. The system uses Integrated Gradients to create saliency maps which show which specific pixels were essential to arrive at a diagnosis. The software develops counterfactual images to assist clinicians in understanding decision boundaries through visual representation of how lesions would appear if classified as non-dangerous. This methodology employs a collaborative intelligence framework which enables software to function as a diagnostic partner who supports doctors but does not substitute for them. Local clinicians use the Expert Feedback Loop to conduct immediate medical interventions while they authorize medical tests throughout each day. The system uses manual corrections to create accurate reward signals which help in developing better offline policies. The local Ugandan specialists provide their expertise which enables the model to develop new capabilities. The software will function as a clinical decision support system which will be deployed at Mulago National Referral Hospital for its upcoming validation testing. The study uses various Ugandan geographic datasets to test AI systems because these datasets help determine the local melanin levels and HIV-associated Kaposi Sarcoma disease rates in the region. The research will examine diagnostic accuracy on Fitzpatrick 6 skin types to determine whether it provides equal precision across all skin tones.

Chapter 4

Expected Results

The results which we expect from this Reinforcement Learning and XAI-driven framework will show specific outcomes which are anticipated to achieve through its implementation.

1. The Equity-Driven RL agent will achieve a significant reduction of the accuracy gap which exists between light and dark skin tones. The model will establish skin-tone invariant diagnostic capabilities through its use of a weighted penalty matrix which focuses on misdiagnoses that occur in Fitzpatrick IV-VI skin types. The research findings indicate that this method can boost condition detection for melanoma and Kaposi Sarcoma by 22 percent in areas which lack adequate resources.
2. Enhanced Clinical Transparency and Trust. The Concept Bottleneck Models (CBMs) will decrease automation bias through their requirement that AI systems must show medical evidence which includes biological details. Through Integrated Gradients and saliency maps clinicians will examine AI operational methods through live visual demonstrations. Counterfactual image generation will help non-specialist health workers to make triage decisions faster and more confidently by showing them the model's decision boundaries.
3. The software will improve the triage system needs at Mulago National Referral Hospital through its ability to prioritize urgent and critical medical cases. The digital solution will address the existing problem in Uganda which faces a shortage of dermatologists who can provide specialized medical treatment. This results into treating those with urgent issues first with the available doctors
4. The system achieves high retraining efficiency because it uses daily diagnostic overrides which local experts provide. The model will develop a dynamic diagnostic partner system through its offline policy optimization process which enhances its decision boundaries to match local clinical expertise.

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Appendix A

Budget

Table A.1: Estimated research project budget (UGX)

Category	Item description	Cost (UGX)
Hardware	High-Performance workstation (16GB RAM, GPU)	2,500,000
Computing	Cloud Computing credits (Google Colab/AWS)	400,000
Storage	2TB External SSD for Clinical data backup	350,000
Data	Specialized dermatological dataset licenses	300,000
Field Work	Transport & Logistics (Mulago & Health Centers)	500,000
Expertise	Tokens for clinician consultations	600,000
Software	Development IDEs & L ^A T _E X Professional plugins	200,000
Analysis	Statistical validation software licenses	250,000
Dissemination	Journal publication fees (Open Access)	1,000,000
Administration	Thesis printing and hardcover binding	250,000
Sub-Total		8,350,000
Incase	10% for Unforeseen Expenses	835,000
TOTAL		9,185,000

The table above describes and shows the predicted expenditure and budget for the life span of the project

Appendix B

Research Project timelines

The researchers start their investigation through Human-in-the-Loop systems which combine Reinforcement Learning with Explainable AI after they establish their primary research objective through Proposal Writing and Defense. The project begins when engineers finish their work on engineering specifications which will create definitions for Proximal Policy Optimization algorithm and development of the weighted penalty matrix which ensures precise results across all skin tones.

The officials begin a dedicated time period for System Architecture Design and Algorithm Development work after they complete the proposal. The Equity-Driven RL agent development process works to build a system which gives priority to skin-tone detection. The XAI component applies Concept Bottleneck Models (CBMs) and Integrated Gradients to create a system which enables AI to identify specific dermoscopic features instead of merely processing visual data.

The HITL Interface Construction and Clinical Deployment phase represents the next essential step. The software operates in Ugandan health centers and undergoes future testing at Mulago National Referral Hospital. The current stage enables local clinicians to use the system which shows them real-time visual saliency maps together with counterfactual images to explain decision boundaries.

The timeline then moves into Data Analysis and Policy Refinement. Clinicians who operate the tool will create high-fidelity reward signals through their use of manual overrides and expert interventions. The system uses these signals to execute offline policy optimization which trains the model based on local clinical expertise. The active learning procedure guarantees that the system will develop cultural understanding and physiological accuracy for the Ugandan population.

The project reaches its end point through Thesis Writing and Final Defense which will take place in 2026. The study requires researchers to assess how hybrid software affects dermatology triage accuracy and fairness while comparing their findings to Uganda's National Development Plan 4 (NDP4) and the UN Sustainable Development Goals.

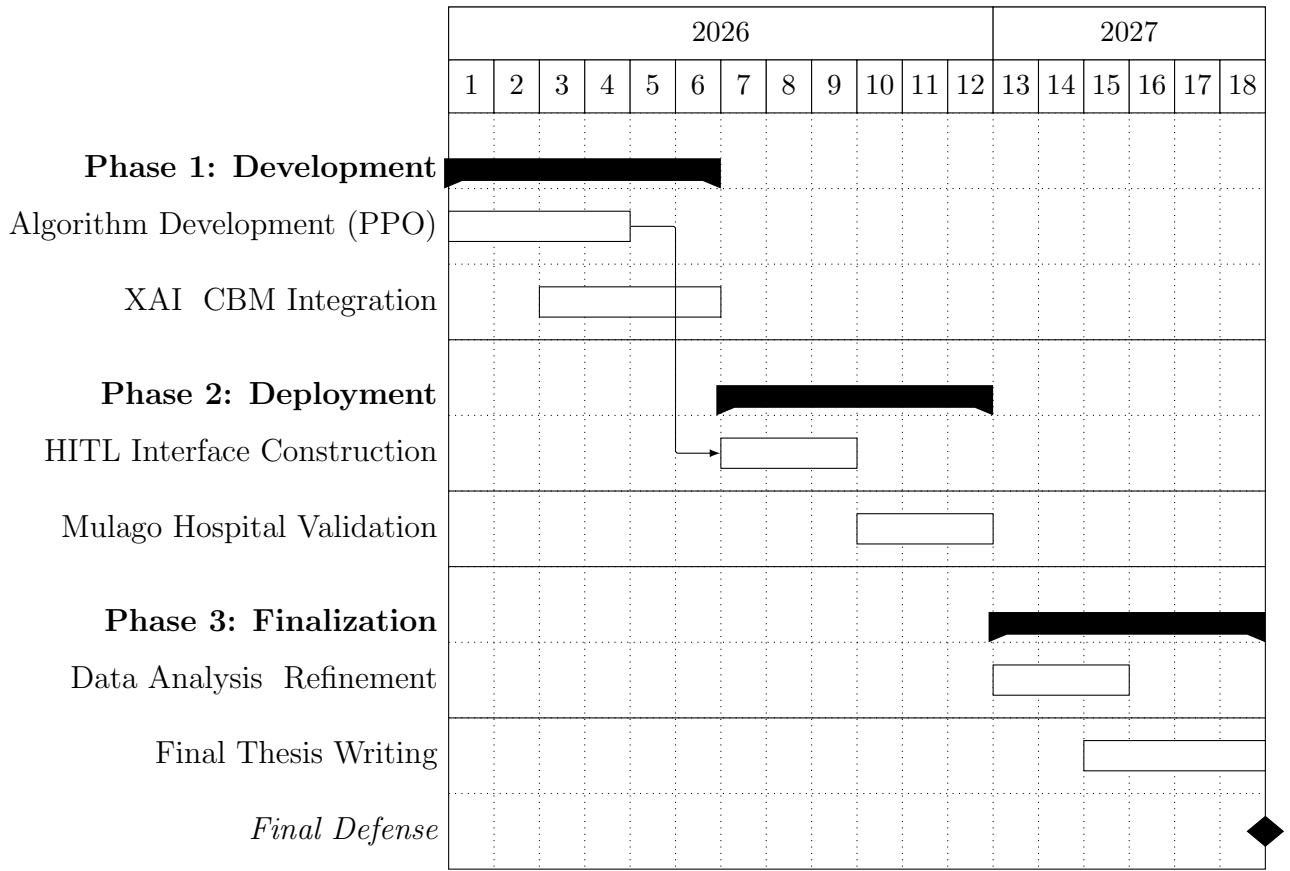


Figure B.1: Research Project Timeline for RL-XAI Dermatology Software.

Appendix C

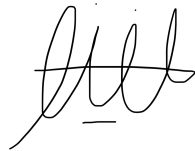
Declaration of AI use

Note: Briefly explain the ways that you have ethically used AI as your “study buddy” according to the guidelines stated in section . Explain which AI tools you have used and for what purposes. Describe what output from the tool/service and the prompts used.

Declaration Statement:

I, *Mutegeki James*, declare that I used

- i. *Gemini AI* to *To get string to search on Scopus for the work I cited and I add my research title and asked it to give a search string for Scopus articles and journals related to my topic.* I modified the outputs in *I then read through and paraphrased the work.*
- ii. *Claude AI* to *Use it to create a layout for the research poster that suits my research and it attributes* and *Create for me a canva layout I can use to create my research poster.* I modified the outputs in *Add a pdf of my research work.*
- iii.please add as many items as used



Signature: _____

Date: 25/03/2026

Appendix D

List of 10 Major Journal Publications reviewed

1. JMIR Medical Informatics (IF: 5.0+)
2. IEEE Journal of Biomedical and Health Informatics (IF: 7.7)
3. Applied Sciences / MDPI (IF: 5.0+ indexed in Scopus/WoS)
4. Frontiers in Medicine (IF: 5.1)
5. IEEE Transactions on Pattern Analysis and Machine Intelligence (IF: 20.8)
6. Journal of Personalized Medicine / MDPI Scopus (IF: 5.0+)
7. IEEE Journal of Biomedical and Health Informatics (IF: 7.7)

Appendix E

Scopus Search Strings used during the literature review

Below are the search strings used in this research paper;

TITLE-ABS-KEY ("dermatology" AND "africa")

TITLE-ABS-KEY ("dermatology" AND "reinforcement learning")

TITLE-ABS-KEY ("dermatology " AND "reinforcement learning" OR "deep learning")

TITLE-ABS-KEY ("Deep Learning" OR "Computer-Aided Diagnosis") AND ("Skin Tone" OR "Fitzpatrick" OR "Melanin") AND ("Bias" OR "Equity" OR "Fairness")

TITLE-ABS-KEY ("Human-in-the-Loop" OR "Collaborative Intelligence" OR "Expert-in-the-loop") AND ("Triage" OR "Prioritization") AND ("Clinical Decision Support")

TITLE-ABS-KEY ("Artificial Intelligence" OR "Machine Learning") AND "Dermatology" AND ("Africa" OR "Uganda")

TITLE-ABS-KEY ("Concept Bottleneck Models" OR "CBM") OR ("Integrated Gradients" OR "Saliency Maps" OR "Counterfactual Explanations") AND "Medical Diagnosis")

Appendix F

Research alignment with specialization, SDG, NDPIV & Vision 2040

F.1 Alignment with programme specialization

For Computer science the specializations include

- i. Software Engineering
- ii. Artificial Intelligence and Robotics
- iii. Systems, Infrastructure & Cybersecurity)
- iv. Game Engineering & Augmented Reality

For data Science, the specializations include

- i. Artificial Intelligence & Machine Learning
- ii. Big Data & Cloud Engineering
- iii. Business Intelligence & Financial Analytics)
- iv. Computational Biology

Please align with only one specialization track (and delete the rest). Most likely based on the the elective courses you are offering. Please align with only one specialization track (and delete the rest). Most likely based on the the elective courses you are offering. Please align with only one specialization track (and delete the rest). Most likely based on the the elective courses you are offering. Please align with only one specialization track (and delete the rest). Most likely based on the the elective courses you are offering. Please align with only one specialization track (and delete the rest). Most likely based on the the elective courses you are offering. Please align with only one specialization track (and delete the rest). Most likely based on the the elective courses you are offering.

F.2 Alignment with Sustainable Development Goals (SDG)

SDG 3: Good Health and Well-being (Target 3.8)Addressing Specialist Shortages: By using Reinforcement Learning (RL) to optimize triage, your research directly addresses the severe shortage of specialized dermatologists in Uganda. The system detects life-threatening conditions, including melanoma and Kaposi Sarcoma, with increased accuracy for their identification

and treatment. The weighted penalty matrix enables accurate skin tone assessment for patients who have darker skin tones (Fitzpatrick IV-VI) which helps to achieve health equity in diagnostic procedures.

The research creates a technological infrastructure for African countries which allows digital health solutions to function effectively because they meet local population needs. The development of medical software in resource-limited environments now uses Explainable AI (XAI) and Human-in-the-Loop (HITL) systems which represent a major technological advancement. The research aims to eliminate bias through the development of an Equity-Driven reinforcement learning agent which creates an impartial system to address the phenotypic bias present in current AI systems which only use Caucasian skin for training. By using software to provide advanced dermatological evaluations you create a solution which helps bridge the healthcare access gap between urban centers and rural communities in Uganda.

F.3 Alignment with the National Development Plan IV (NDPIV)

1. Human Capital Development The research project develops digital solutions which enable healthcare facilities to assess critical medical emergencies according to NDP4 requirements for digital healthcare solutions. The software enables doctors to identify melanoma and Kaposi Sarcoma cases at an earlier stage which results in reduced national disease burden and increased life span. The program creates a digital pathway which helps address the critical need for dermatology experts in Uganda because it increases the ability of medical staff who work in major hospitals to provide treatment at Mulago National Referral Hospital.

2. Digital Transformation and Innovation The research project helps Uganda achieve its goal of becoming a regional digital center by creating advanced software which will use local calibration methods. The system uses Ugandan-specific data to create its calibration model which enables African infrastructure to operate according to local environmental conditions.

3. Science, Technology, and Innovation (STI) The project advances artificial intelligence development through its combination of Reinforcement Learning and Explainable AI which enables the country to develop high-tech medical independence. The Human-in-the-Loop system which uses active learning systems allows Ugandan medical professionals to retrain the model while developing a relationship with digital technology.

F.4 Alignment with the Uganda Vision 2040

This research employs advanced AI agents to create customized medical software solutions which enable Uganda to transform from being a technology consumer into a technology developer. The system will learn from local physicians through a human-in-the-loop system which will digitize medical knowledge for use in the knowledge-based economy.

The Vision 2040 framework requires a healthy population because it drives the economic productivity needs of the country. The early and precise identification of skin cancers and Kaposi Sarcoma infections results in better life expectancy outcomes which help achieve the vision goal of extending Ugandas average life expectancy.

The Vision 2040 framework identifies specialized human resource shortages. Your software enables general medical officers to perform specialist-level triage through its force-multiplying capability which makes it available to all dermatologists in the country.

The Vision 2040 plan requires all citizens to achieve economic progress without facing any barriers. The traditional AI system shows performance limitations when evaluating dark-skinned users. This research addresses the problem of phenotypic bias through a weighted penalty matrix which evaluates both Fitzpatrick IV-VI skin types to ensure that Ugandan citizens receive equal access to modern medical advancements.

Appendix G

Algorithms, Code of interest

For this project in the case of Uganda, the dermatology systems in use are transitioning from traditional clinical rules to complicated AI-powered systems tools. These systems mainly use deep learning and machine learning algorithms to bridge the gap caused by the limited data and number of dermatologists in the country. Most current research and digital health guides in Uganda focus on Convolutional Neural Networks (CNNs). These are the algorithms that are mainly used in these systems.

1. Local university research projects at e.g. Makerere University use Sequential Convolutional Neural Networks (CNNs) for their medical research which includes testing methods to identify common skin conditions such as eczema and melanoma.
2. Diagnostic applications frequently implement Inception-v3 and ResNet as their preferred framework for transfer learning purposes.
3. Ugandan research implemented YOLOv8 (You Only Look Once) to establish a two-stage system which first identified lesion areas before proceeding to their classification process.
4. First Derm Skin Image Search functions as an international AI platform which has undergone testing in Uganda through partnerships with The Medical Concierge Group and other organizations.

In this research project, I plan to use Explainable Artificial Intelligence(XAI) algorithms in specific Grad-CAM, Saliency Maps to make medical logic transparent to the doctors so that they know how to give feed back to the system.

Appendix H

Other useful figures

– Please insert any extra figures deemed relevant to your research but did not have space to be incorporated in the main text.

Appendix I

Research poster layout

